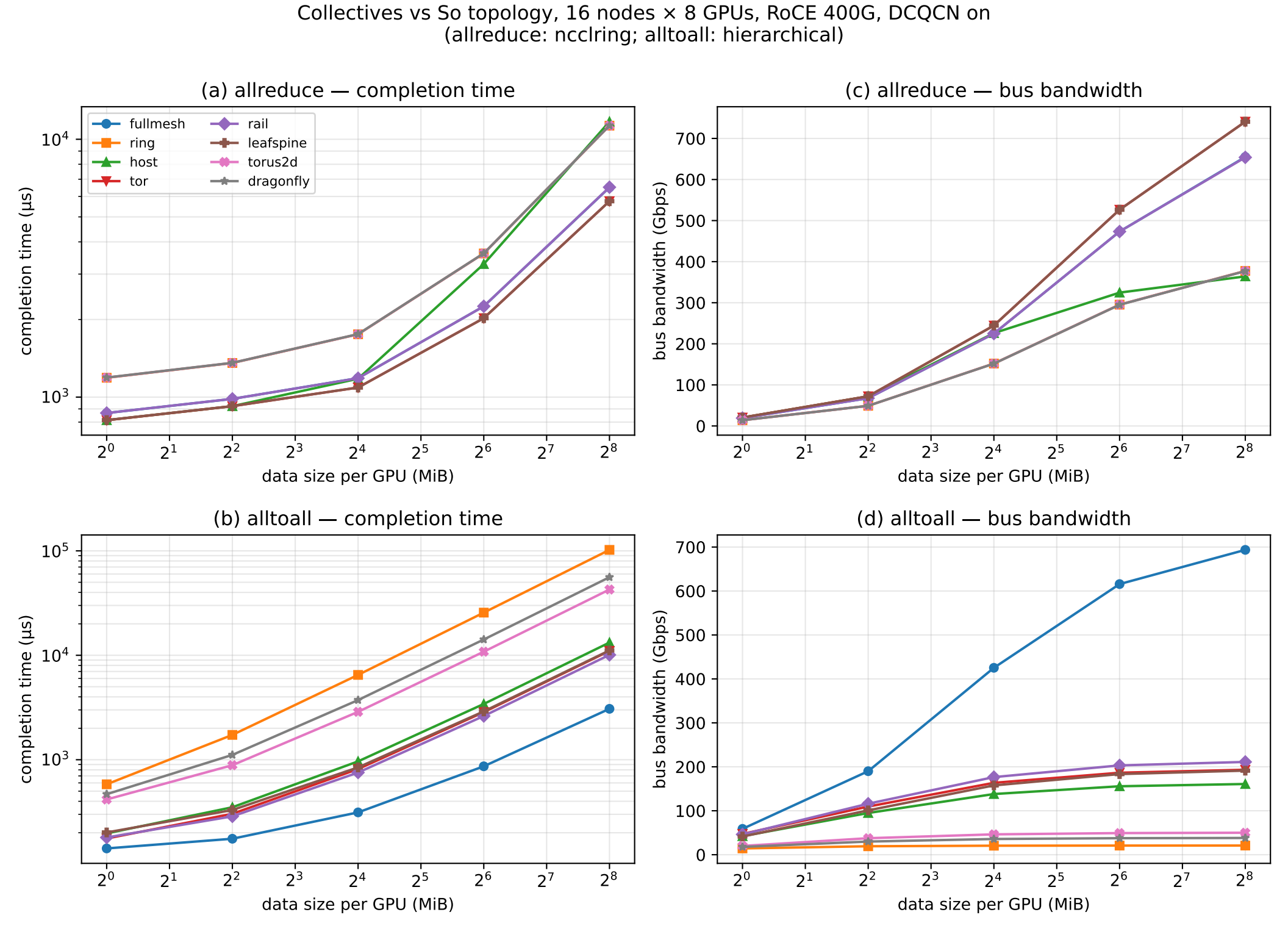
Collectives vs So topology, 16 nodes × 8 GPUs, RoCE 400G, DCQCN on
(allreduce: ncclring; alltoall: hierarchical)
(a) allreduce — completion time
(b) alltoall — completion time
(c) allreduce — bus bandwidth
(d) alltoall — bus bandwidth
fullmesh
ring
host
tor
rail
leafspine
torus2d
dragonfly
completion time (µs)
bus bandwidth (Gbps)
data size per GPU (MiB)